

**WESTERN SYDNEY**  
UNIVERSITY



**School of Computer, Data and Mathematical Sciences**



## Subject Outline

COMP1013 (Vietnam) Analytics Programming  
Vietnam Operation Trimester 2 2026

Western Sydney University acknowledges the peoples of the Darug, Dharawal, Eora and Wiradjuri nations. We also acknowledge that the teaching and learning currently delivered across our campuses is a continuance of the teaching and learning that has occurred on these lands for tens of thousands of years.

## Subject Details

|                           |                                                            |
|---------------------------|------------------------------------------------------------|
| <b>Subject Code:</b>      | COMP1013 (Vietnam)                                         |
| <b>Subject Name:</b>      | Analytics Programming                                      |
| <b>Credit Points:</b>     | 10                                                         |
| <b>Subject Level:</b>     | 1                                                          |
| <b>Assumed Knowledge:</b> | Familiarity with computer software programs such as Excel. |

Note: Students with any problems, concerns or doubts should discuss those with the Subject Coordinator as early as they can.

## Subject Coordinator

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# 1 About Analytics Programming

## 1.1 An Introduction to this Subject

This subject is a practical introduction to coding and problem-solving in data science using the R programming language. You'll learn to develop and analyse R scripts to perform simple data analysis with diverse data sets. The subject also teaches you to create professional, reproducible reports with visualisations and introduces you to industry-standard tools like Linux and Git. By the end, you'll have a strong foundation in data, technology, and computer science.

## 1.2 What is Expected of You

### Study Load

A student is expected to study an hour per credit point a week. For example a 10 credit point subject would require 10 hours of study per week. This time includes the time spent within classes during lectures, tutorials or practicals.

*Note for Summer Terms: As Summer subjects deliver the same content and classes over a shorter period of time, the subjects are run in a more intensive mode. Regardless of the delivery mode, the study hours for each subject in Summer will be around 30 hours.*

### Attendance

Lectures and tutorials are designed to scaffold learning and assist students to gain an understanding of the subject content, and subsequently to complete their assessment tasks. Students should note that attendance is considered critical for students to gain full value from this subject, and is closely correlated with student success and grades. Students are encouraged to participate in all class activities, as non-engagement may seriously undermine a student's ability to satisfactorily complete the subject.

## Approach to Learning

### Lecture

2 hour face to face lectures. These sessions will be used to introduce concepts and work through examples. Students are expected to attend all lectures, taking their own notes in addition to following the PDF of the slides provided on vUWS. Students are encouraged to ask questions during and after the lecture.

### Tutorials

2 hour tutorial sessions. These sessions will be used to reinforce concepts introduced in lecture sessions. Tutorial sessions will be held in computer labs that will allow the use of an R programming GUI for a variety of data science problems. Students are expected to attend all tutorial sessions, attempting the questions (posted on vUWS) and asking the tutorial supervisor for assistance. Students do not need to pre-prepare for tutorials.

### Online Learning Requirements

Subject materials will be made available on the subject's vUWS (E-Learning) site (<https://vuws.westernsydney.edu.au/>). You are expected to consult vUWS at least twice a week, as all subject announcements will be made via vUWS. Teaching and learning materials will be regularly updated and posted online by the teaching team.

### Special Requirements

*Essential Equipment:*

Students require access to a computer.

*Legislative Pre-Requisites:*

Not Applicable

### **Online Learning Requirements**

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### **Special Requirements**

*Essential Equipment:*

Students require access to a computer.

*Legislative Pre-Requisites:*

Not Applicable

## **1.3 Changes to Subject as a Result of Past Student Feedback**

The University values student feedback in order to improve the quality of its educational programs. The feedback provided helps us improve teaching methods and subjects of study. The survey subjects results inform subject content and design, Subject Outlines, teaching methods, assessment processes and teaching materials.

You are welcome to provide feedback that is related to the teaching of this subject. At the end of the semester you will be given the opportunity to complete a Student Feedback on Subject questionnaire to assess the subject. If requested by your subject coordinator, you may also have the opportunity to complete a Student Feedback on Teaching (SFT) questionnaire to provide feedback for individual teaching staff.

As a result of student feedback, the following changes and improvements to this Subject have recently been made:

- The structure of the subject has been adjusted to focus on programming

## 2 Learning and Teaching Activities

| Teaching Weeks        |                               | Assessment Due   |
|-----------------------|-------------------------------|------------------|
| Week 1<br>11-05-2026  | Introduction to R Programming |                  |
| Week 2<br>18-05-2026  | R fundamentals                |                  |
| Week 3<br>25-05-2026  | Flow Control                  |                  |
| Week 4<br>01-06-2026  | Data Structure                | - Online Quizzes |
| Week 5<br>08-06-2026  | Functions                     |                  |
| Week 6<br>15-06-2026  | Data                          | - Online Quizzes |
| Week 7<br>22-06-2026  | Literate Programming          | - Online Quizzes |
| Week 8<br>29-06-2026  | Unix shell                    | - Online Quizzes |
| Week 9<br>06-07-2026  | Version control               |                  |
| Week 10<br>13-07-2026 | Visualisation                 | - Online Quizzes |
| Week 11<br>20-07-2026 | Relational Data               | - Assignment     |
| Week 12<br>27-07-2026 | Revision                      |                  |
| Week 13<br>03-08-2026 |                               | - Final Exam     |

The above timetable should be used as a guide only, as it is subject to change. Students will be advised of any changes as they become known on the Subject's vUWS site.

## 3 Assessment Information

### 3.1 Subject Learning Outcomes

|   | Outcome                                                                                                                          |
|---|----------------------------------------------------------------------------------------------------------------------------------|
| 1 | Write R programs to conduct Data Science tasks.                                                                                  |
| 2 | Debug programs to fix R scripts that perform data science and analytics tasks.                                                   |
| 3 | Create basic data visualisations and reproducible reports using R graphics, Markdown, and R Markdown to support decision making. |
| 4 | Integrate large data sets to enable complex data analysis.                                                                       |
| 5 | Manage script development through version control and testing in the Git environment with the help of Unix commands.             |

### 3.2 Assessment Summary

The assessment items in this subject are designed to enable you to demonstrate that you have achieved the subject learning outcomes. Completion and submission of all assessment items which have been designated as mandatory or compulsory is essential to receive a passing grade.

**To pass this subject you must:**

- Attempt all assessment tasks
- Achieve at least 50% overall
- score at least 50% (threshold) on the exam

| Item           | Weight | Due Date                         | SLOs Assessed | Mandatory | Threshold |
|----------------|--------|----------------------------------|---------------|-----------|-----------|
| Online Quizzes | 20%    | Weeks 4, 6, 7, 8, and 10         | 1, 2, 3       | Yes       | No        |
| Assignment     | 40%    | Week 11                          | 1, 2, 3, 4, 5 | Yes       | No        |
| Final Exam     | 40%    | During formal examination period | 2, 3, 4, 5    | Yes       | Yes       |

### Feedback on Assessment

Feedback is an important part of the learning process that can improve your progress towards achieving the learning outcomes. Feedback is any written or spoken response made in relation to academic work such as an assessment task, a performance or product. It can be given to you by a teacher, an external assessor or student peer, and may be given individually or to a group of students. As a Western Sydney University student, it is your responsibility to seek out and act on feedback that is provided to you as a resource to further your learning.

In this subject, you can expect verbal and/or written feedback within 2 weeks of any assessment submission. Further informal feedback will also be provided in lectures and/or tutorial sessions.

### Academic Integrity and Student Misconduct Rule

Western cares about your success as a student and in your future career. Studying with academic integrity safeguards your professional reputation and your degree. All Western students must:

- be familiar with the policies listed below;
- apply principles of academic integrity;
- act honestly and ethically in producing all academic work and assessment tasks; and
- submit work that is their own and acknowledge any sources used in their work.

Each time you submit an assessment, you will declare that you have completed it individually, unless it is a group assignment. In the case of a group assignment, each group member should be ready to document their individual contribution if needed.

The [Student Misconduct Rule](#) applies to all students of Western Sydney University including Western Sydney University programs taught by other education providers. You must not engage in academic, research or general misconduct as defined in the Rule or you may be subject to sanctions. The University considers submitting falsified documentation in support of requests to redo, resit or extend submissions, including sitting of deferred examinations, as instances of general misconduct.

More information is available in the [Academic Integrity Guidelines](#). It is your responsibility to apply these principles to all work you submit to the University.

### **Special Consideration and Requests for Extensions**

Western recognises that there may be times when things outside of your control impact your ability to complete your studies.

You can submit a Special Consideration or extension application to request that you are:

- granted an extension,
- excused from a compulsory teaching activity,
- being offered an opportunity to complete a different, academically equivalent assessment task or
- granted another appropriate adjustment.

Before submitting your application, you should:

- gather any supporting documentation that demonstrates how you have been impacted by circumstances outside of your control; and
- refer to the [Supporting Documentation](#) website for guidance on acceptable evidence.

If you do not have documentation at the time of submitting your application, you should still apply. You may be asked to provide supporting documents at a later stage.

### **Need help?**

If you are having difficulties with understanding or completing an assessment task, contact your Subject Coordinator as soon as possible. Western also has a range of academic support services, including:

- **Library Study Smart:** book a one-to-one [Zoom consultation](#) with a literacy expert. You can discuss how to develop your assignment writing and study skills or seek assistance to [understand referencing and citation requirements](#). Check the Library Study Smart website for how-to [study guides and tools](#).
- **Studiosity:** Upload your assignment draft to Studiosity within vUWS to receive writing feedback within 24 hours.
- **Online workshops, programs and resources:** From maths and stats help to academic literacy and peer support programs, the University has a range of resources to assist.

Please also remember that there is a range of [wellbeing support available](#) - from counselling and disability services to welfare.

### 3.2.1 Online Quizzes

|                                        |                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Weight:</b>                         | 20%                                                                                                         |
| <b>Type of Collaboration:</b>          | Individual                                                                                                  |
| <b>Due:</b>                            | Weeks 4, 6, 7, 8, and 10                                                                                    |
| <b>Submission:</b>                     | Online through vUWS                                                                                         |
| <b>Format:</b>                         | 30 minutes online through vUWS or other means. The quizzes will be completed in class during the tutorials. |
| <b>Length:</b>                         | 5x 30 minutes                                                                                               |
| <b>Use of Artificial Intelligence:</b> | No.                                                                                                         |

#### Instructions:

The quiz will be based on the material covered in lectures and practicals. It may consist of multiple choice or short answer or numeric questions. Some questions may require the use of software to compute a solution.

The quizzes will be completed in class during the tutorials

**Marking Criteria:**

| Criteria | High Distinction | Distinction  | Credit       | Pass         | Unsatisfactory |
|----------|------------------|--------------|--------------|--------------|----------------|
| Mark     | at least 85%     | at least 75% | at least 65% | at least 50% | less than 50%  |

### 3.2.2 Assignment

|                                        |                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Weight:</b>                         | 40%                                                                                                        |
| <b>Type of Collaboration:</b>          | Individual                                                                                                 |
| <b>Due:</b>                            | Week 11                                                                                                    |
| <b>Submission:</b>                     | online through vUWS, and code submitted via git repo.                                                      |
| <b>Format:</b>                         | Written report with results. The code needs to be submitted via git, repository details to be given later. |
| <b>Length:</b>                         | Approx. 2000 words                                                                                         |
| <b>Use of Artificial Intelligence:</b> | No.                                                                                                        |

#### Instructions:

The Assignment will consist of practical activities and a written report.

The code is to be submitted via git repository to gauge GIT usage, details to be provided later.

**Marking Criteria:**

| Criteria | High Distinction | Distinction  | Credit       | Pass         | Unsatisfactory |
|----------|------------------|--------------|--------------|--------------|----------------|
| Mark     | at least 85%     | at least 75% | at least 65% | at least 50% | less than 50%  |

### 3.2.3 Final Exam

|                                        |                                                                    |
|----------------------------------------|--------------------------------------------------------------------|
| <b>Weight:</b>                         | 40%                                                                |
| <b>Type of Collaboration:</b>          | Individual                                                         |
| <b>Due:</b>                            | During formal examination period                                   |
| <b>Submission:</b>                     | via vUWS                                                           |
| <b>Format:</b>                         | 1-hour exam using a computer in the exam room.                     |
| <b>Length:</b>                         | 10-15 minutes oral exam. 1 hour written exam for TNE campuses only |
| <b>Threshold Detail:</b>               | 50%                                                                |
| <b>Use of Artificial Intelligence:</b> | No.                                                                |

#### Instructions:

The exam will consist of practical based questions using computer software and data sets.

Detailed instruction of the exam will be provided later in the semester via vUWS

**Marking Criteria:**

| Criteria         | High Distinction                                                                                                                                       | Distinction                                                                                                     | Credit                                                                                                                 | Pass                                                                                           | Unsatisfactory                                                                               |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Marking criteria | Demonstrating clear , complete and comprehensive understanding of the subject content; with development of relevant skills to an extremely high level. | Demonstrating clear understanding of most subject content; with development of relevant skills to a high level. | Demonstrting understanding of some major subject content; with development of relevant skills to a satisfactory level. | Demonstrating understading of a range of subject content; with developemnt of relevant skills. | Demonstrating no understanding of basic subject content, and no relevant skills development. |

## 4 Readings and Resources

*You can find all required readings in the Reading List section of this subject's site on the Learning Management System.*

## 5 Key Teaching and Learning Policies

The University has several policies that relate to teaching and learning. Links to important policies affecting students are below. It is your responsibility to ensure you familiarise yourself with these policies so that you are aware of your rights and responsibilities.

- [Assessment Policy](#)
- [Assessment Procedures - Review of Grade](#)
- [Bullying Prevention Policy](#)
- [Special Consideration Policy](#)
- [Student Administration Policy](#)
- [Examinations Policy](#)
- [Learning and Teaching Policy](#)
- [Progression Procedure](#)
- [Student Code of Conduct](#)
- [Student Misconduct Rule](#)